

SQL PROJECT

INDIA GENERAL ELECTION RESULT ANALYSIS 2024

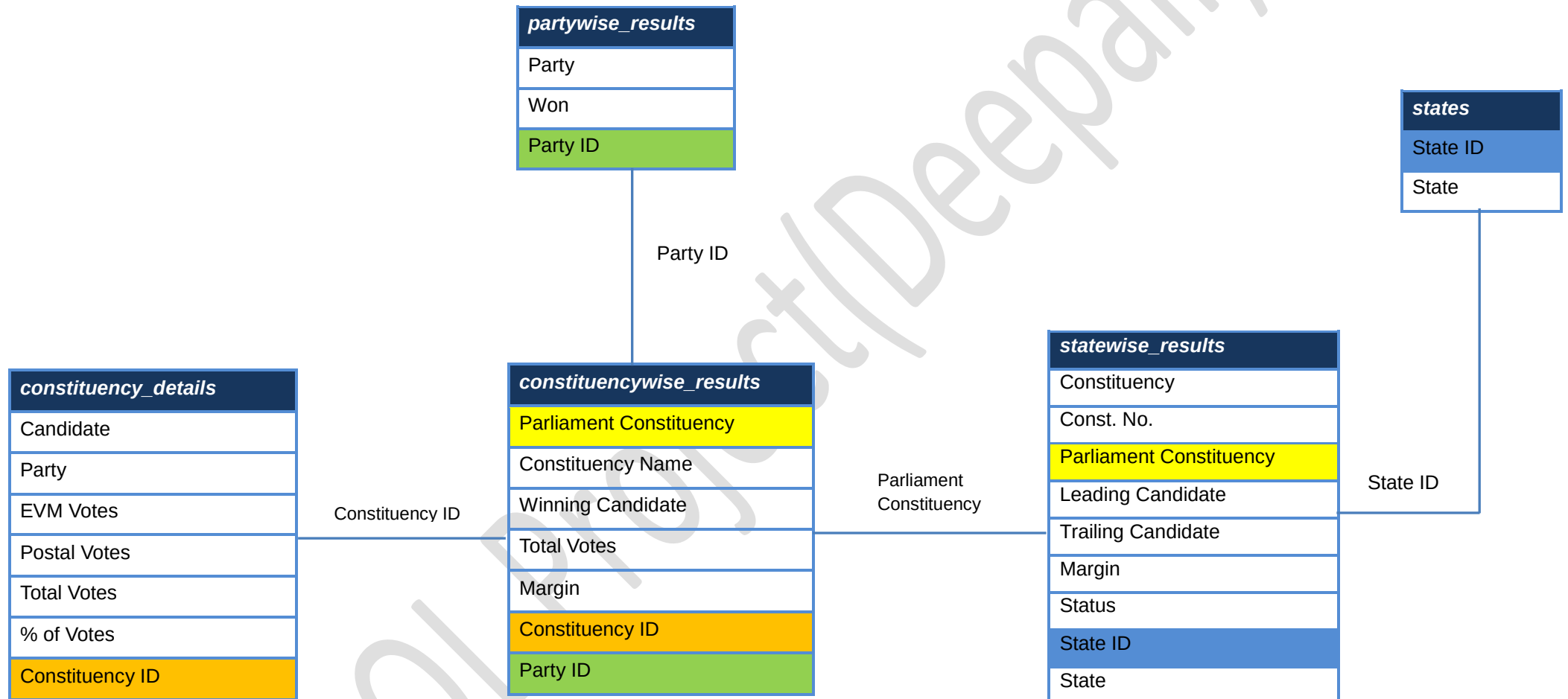
PROBLEM STATEMENT

- Total Seats.
- What are the total numbers of seats available for elections in each state?
- Total Seats Won by NDA Alliance.
- Seats Won by NDA Alliance Parties.
- Total Seats Won by I.N.D.I.A. Alliance.
- Seats Won by I.N.D.I.A. Alliance Parties.
- Add new column field in table partywise_results to get the Party Allianz as NDA, I.N.D.I.A. and OTHER
- Which party alliance (NDA, I.N.D.I.A. or OTHER) won the most seats across all states?
- Winning candidates name, their party name, total votes, and the margin of victory for a specific state and constituency?
- What is the distribution of EVM votes versus postal votes for candidates in a specific constituency?
- Which parties won the most seats in s State, and how many seats did each party win?
- What are the total numbers of seats won by each party alliance (NDA, I.N.D.I.A. and OTHER) in each state for the India Elections 2024?
- Which candidate received the highest number of EVM votes of each constituency (Top 10)?
- Which candidate won and which candidate was the runner-up in each constituency of State for the 2024 elections?
- For the state of Maharashtra, What are the total number of seats, total number of candidates, total number of parties, total votes (including EVM and postal), and the breakdown of EVM and postal votes?

FUNCTIONALITIES I USED

- ✓ DISTINCT
- ✓ COUNT()
- ✓ SUM()
- ✓ CASE
- ✓ JOIN
- ✓ ON
- ✓ ORDER BY
- ✓ GROUP BY
- ✓ MAX ()
- ✓ ROW_NUMBER
- ✓ SELECT
- ✓ PARTITION BY
- ✓ OVER()
- ✓ WITH
- ✓ UPDATE
- ✓ ALTER TABLE
- ✓ ADD
- ✓ SET
- ✓ WHERE
- ✓ THEN
- ✓ ELSE
- ✓ HAVING
- ✓ WINDOW ()
- ✓ CTE
- ✓ AVG ()
- ✓ RANK
- ✓ DATE FUNCTIONS
- ✓ INNER JOIN
- ✓ LEFT JOIN
- ✓ MULTI TABLE JOINS
- ✓ SCHEMAS
- ✓ IIF ()
- ✓ CAST()

ENTITY RELATIONSHIP DIAGRAM (ERD) / SCHEMA



QUERY FOR PROBLEM STATEMENTS

1. Total Seats.

```
SELECT DISTINCT COUNT (Parliament_Constituency) AS Total_Seats
FROM constituencywise_results
```

100 % ✓ No issues found

Results Messages

	Total_Seats
1	543

2. What are the total numbers of seats available for elections in each state?

```
SELECT s.State AS State_Name, COUNT(cr.Constituency_ID) AS Total_Seats_Available
FROM constituencywise_results cr
JOIN statewise_results sr ON cr.Parliament_Constituency = sr.Parliament_Constituency
JOIN states s ON sr.State_ID = s.State_ID
GROUP BY s.State
ORDER BY s.State
```

100 % ✓ No issues found

Results Messages

	State_Name	Total_Seats_Available
1	Andaman & Nicobar Islands	1
2	Andhra Pradesh	25
3	Arunachal Pradesh	2
4	Assam	14
5	Bihar	40
6	Chandigarh	1
7	Chhattisgarh	11
8	Dadra & Nagar Haveli and Daman & Diu	2
9	Delhi	7
10	Goa	2
11	Gujarat	26
12	Haryana	10
13	Himachal Pradesh	4
14	Jammu and Kashmir	5
15	Jharkhand	14

16	Karnataka	28
17	Kerala	20
18	Ladakh	1
19	Lakshadweep	1
20	Madhya Pradesh	29
21	Maharashtra	48
22	Manipur	2
23	Meghalaya	2
24	Mizoram	1
25	Nagaland	1
26	Odisha	21
27	Puducherry	1
28	Punjab	13
29	Rajasthan	25
30	Sikkim	1
31	Tamil Nadu	39
32	Telangana	17
33	Tripura	2
34	Uttar Pradesh	80
35	Uttarakhand	5
36	West Bengal	42

3. Total Seats Won by NDA Alliance.

```
SELECT SUM (CASE
    WHEN party IN (
        'Bharatiya Janata Party - BJP',
        'Telugu Desam - TDP',
        'Janata Dal (United) - JD(U)',
        'Shiv Sena - SHS',
        'AJSU Party - AJSUP',
        'Apna Dal (Soneylal) - ADAL',
        'Asom Gana Parishad - AGP',
        'Hindustani Awam Morcha (Secular) - HAMS',
        'Janasena Party - JnP',
        'Janata Dal (Secular) - JD(S)',
        'Lok Janshakti Party(Ram Vilas) - LJPRV',
        'Nationalist Congress Party - NCP',
        'Rashtriya Lok Dal - RLD',
        'Sikkim Krantikari Morcha - SKM') THEN [Won]
    ELSE 0 END) AS NDA_Total_Seats_Won
FROM partywise_results
```

100 % No issues found

Results	
Messages	
NDA_Total_Seats_Won	
1	292

4. Seats Won by NDA Alliance Parties.

```
SELECT party as Party_Name, won as Seats_Won FROM partywise_results WHERE party IN (
    'Bharatiya Janata Party - BJP',
    'Telugu Desam - TDP',
    'Janata Dal (United) - JD(U)',
    'Shiv Sena - SHS',
    'AJSU Party - AJSUP',
    'Apna Dal (Soneylal) - ADAL',
    'Asom Gana Parishad - AGP',
    'Hindustani Awam Morcha (Secular) - HAMS',
    'Janasena Party - JnP',
    'Janata Dal (Secular) - JD(S)',
    'Lok Janshakti Party(Ram Vilas) - LJPRV',
    'Nationalist Congress Party - NCP',
    'Rashtriya Lok Dal - RLD',
    'Sikkim Krantikari Morcha - SKM')
ORDER BY Seats_Won DESC
```

100 % ✓ No issues found

Results Messages

	Party_Name	Seats_Won
1	Bharatiya Janata Party - BJP	240
2	Telugu Desam - TDP	16
3	Janata Dal (United) - JD(U)	12
4	Shiv Sena - SHS	7
5	Lok Janshakti Party(Ram Vilas) - LJPRV	5
6	Janata Dal (Secular) - JD(S)	2
7	Janasena Party - JnP	2
8	Rashtriya Lok Dal - RLD	2
9	Sikkim Krantikari Morcha - SKM	1
10	Nationalist Congress Party - NCP	1
11	Hindustani Awam Morcha (Secular) - HAMS	1
12	Asom Gana Parishad - AGP	1
13	AJSU Party - AJSUP	1
14	Apna Dal (Soneylal) - ADAL	1

5. Total Seats Won by I.N.D.I.A. Alliance.

```
SELECT SUM(CASE
    WHEN party IN (
        'Indian National Congress - INC',
        'Aam Aadmi Party - AAP',
        'All India Trinamool Congress - AITC',
        'Bharat Adivasi Party - BHRTADVSIP',
        'Communist Party of India (Marxist) - CPI(M)',
        'Communist Party of India (Marxist-Leninist) (Liberation) - CPI(ML)(L)',
        'Communist Party of India - CPI',
        'Dravida Munnetra Kazhagam - DMK',
        'Indian Union Muslim League - IUML',
        'Nat`Jammu & Kashmir National Conference - JKN',
        'Jharkhand Mukti Morcha - JMM',
        'Jammu & Kashmir National Conference - JKN',
        'Kerala Congress - KEC',
        'Marumalarchi Dravida Munnetra Kazhagam - MDMK',
        'Nationalist Congress Party Sharadchandra Pawar - NCPSP',
        'Rashtriya Janata Dal - RJD',
        'Rashtriya Loktantrik Party - RLTP',
        'Revolutionary Socialist Party - RSP',
        'Samajwadi Party - SP',
```

```

        'Shiv Sena (Uddhav Balasaheb Thackrey) - SHSUBT',
        'Viduthalai Chiruthaigal Katchi - VCK') THEN [Won]
    ELSE 0
END) AS INDIA_Total_Seats_Won
FROM partywise_results

```

100 % No issues found

INDIA_Total_Seats_Won	
1	234

6. Seats Won by I.N.D.I.A. Alliance Parties.

```

SELECT
party as Party_Name,
won as Seats_Won
FROM partywise_results
WHERE party IN (
    'Indian National Congress - INC',
    'Aam Aadmi Party - AAP',
    'All India Trinamool Congress - AITC',
    'Bharat Adivasi Party - BHRTADVSIP',
    'Communist Party of India (Marxist) - CPI(M)',
    'Communist Party of India (Marxist-Leninist) (Liberation) - CPI(ML)(L)',
    'Communist Party of India - CPI',
    'Dravida Munnetra Kazhagam - DMK',
    'Indian Union Muslim League - IUML',
    'Nat`Jammu & Kashmir National Conference - JKN',
    'Jharkhand Mukti Morcha - JMM',
    'Jammu & Kashmir National Conference - JKN',
    'Kerala Congress - KEC',
    'Marumalarchi Dravida Munnetra Kazhagam - MDMK',
    'Nationalist Congress Party Sharadchandra Pawar - NCPSP',
    'Rashtriya Janata Dal - RJD',
    'Rashtriya Loktantrik Party - RLTP',
    'Revolutionary Socialist Party - RSP',
    'Samajwadi Party - SP',
    'Shiv Sena (Uddhav Balasaheb Thackrey) - SHSUBT',
    'Viduthalai Chiruthaigal Katchi - VCK')
ORDER BY Seats_Won DESC

```

100 % ✓ No issues found

Results Messages

	Party_Name	Seats_Won
1	Indian National Congress - INC	99
2	Samajwadi Party - SP	37
3	All India Trinamool Congress - AITC	29
4	Dravida Munnetra Kazhagam - DMK	22
5	Shiv Sena (Uddhav Balasaheb Thackrey) - SHSUBT	9
6	Nationalist Congress Party Sharadchandra Pawar - ...	8
7	Rashtriya Janata Dal - RJD	4
8	Communist Party of India (Marxist) - CPI(M)	4
9	Aam Aadmi Party - AAP	3
10	Indian Union Muslim League - IUML	3
11	Jharkhand Mukti Morcha - JMM	3
12	Jammu & Kashmir National Conference - JKN	2
13	Communist Party of India - CPI	2
14	Communist Party of India (Marxist-Leninist) (Liber...	2
15	Viduthalai Chiruthaigal Katchi - VCK	2
16	Rashtriya Loktantrik Party - RLTP	1
17	Revolutionary Socialist Party - RSP	1
18	Bharat Adivasi Party - BHRTADVSIP	1
19	Kerala Congress - KEC	1
20	Marumalarchi Dravida Munnetra Kazhagam - MDMK	1

7. Add new column field in table partywise_results to get the Party Allianz as NDA, I.N.D.I.A. and OTHER

```
ALTER TABLE partywise_results
ADD party_alliance VARCHAR(50);
```

```
--I.N.D.I.A Allianz
```

```
UPDATE partywise_results
```

```
SET party_alliance = 'I.N.D.I.A.'
```

```
WHERE party IN (
```

```
'Indian National Congress - INC',
```

```
'Aam Aadmi Party - AAP',
```

```
'All India Trinamool Congress - AITC',
```

```
'Bharat Adivasi Party - BHRTADVSIP',
```

```
'Communist Party of India (Marxist) - CPI(M)',
```

```
'Communist Party of India (Marxist-Leninist) (Liberation) - CPI(ML)(L)',  
'Communist Party of India - CPI',  
'Dravida Munnetra Kazhagam - DMK',  
'Indian Union Muslim League - IUML',  
'Jammu & Kashmir National Conference - JKN',  
'Jharkhand Mukti Morcha - JMM',  
'Kerala Congress - KEC',  
'Marumalarchi Dravida Munnetra Kazhagam - MDMK',  
'Nationalist Congress Party Sharadchandra Pawar - NCPSP',  
'Rashtriya Janata Dal - RJD',  
'Rashtriya Loktantrik Party - RLTP',  
'Revolutionary Socialist Party - RSP',  
'Samajwadi Party - SP',  
'Shiv Sena (Uddhav Balasaheb Thackrey) - SHSUBT',  
'Viduthalai Chiruthaigal Katchi - VCK');
```

```
-- NDA Allianz
```

```
UPDATE partywise_results
```

```
SET party_alliance = 'NDA'
```

```
WHERE party IN (
```

```
'Bharatiya Janata Party - BJP',  
'Telugu Desam - TDP',  
'Janata Dal (United) - JD(U)',  
'Shiv Sena - SHS',  
'AJSU Party - AJSUP',  
'Apna Dal (Soneylal) - ADAL',  
'Asom Gana Parishad - AGP',  
'Hindustani Awam Morcha (Secular) - HAMS',  
'Janasena Party - JnP',  
'Janata Dal (Secular) - JD(S)',  
'Lok Janshakti Party(Ram Vilas) - LJPRV',  
'Nationalist Congress Party - NCP',  
'Rashtriya Lok Dal - RLD',  
'Sikkim Krantikari Morcha - SKM');
```

```
-- OTHER
```

```
UPDATE partywise_results
```

```
SET party_alliance = 'OTHER'
```

```
WHERE party_alliance IS NULL;
```

```
SELECT * FROM partywise_results;
```

Results Messages				
	Party	Won	Party_ID	party_alliance
1	Aam Aadmi Party - AAP	3	1	I.N.D.I.A.
2	Asom Gana Parishad - AGP	1	83	NDA
3	All India Majlis-E-Itehadul Muslimeen - AIMIM	1	118	OTHER
4	All India Trinamool Congress - AITC	29	140	I.N.D.I.A.
5	AJSU Party - AJSUP	1	160	NDA
6	Bharatiya Janata Party - BJP	240	369	NDA
7	Communist Party of India - CPI	2	544	I.N.D.I.A.
8	Communist Party of India (Marxist-Leninist) (Li...	2	545	I.N.D.I.A.
9	Communist Party of India (Marxist) - CPI(M)	4	547	I.N.D.I.A.
10	Dravida Munnetra Kazhagam - DMK	22	582	I.N.D.I.A.
11	Hindustani Awam Morcha (Secular) - HAMS	1	664	NDA
12	Indian National Congress - INC	99	742	I.N.D.I.A.
13	Independent - IND	7	743	OTHER
14	Indian Union Muslim League - IUML	3	772	I.N.D.I.A.
15	Janata Dal (Secular) - JD(S)	2	804	NDA
16	Janata Dal (United) - JD(U)	12	805	NDA
17	Jammu & Kashmir National Conference - JKN	2	834	I.N.D.I.A.
18	Jharkhand Mukti Morcha - JMM	3	852	I.N.D.I.A.
19	Janasena Party - JnP	2	860	NDA
20	Kerala Congress - KEC	1	911	I.N.D.I.A.

21	Marumalarchi Dravida Munnetra Kazhagam - ...	1	1046	I.N.D.I.A.
22	Nationalist Congress Party - NCP	1	1142	NDA
23	Rashtriya Janata Dal - RJD	4	1420	I.N.D.I.A.
24	Rashtriya Lok Dal - RLD	2	1458	NDA
25	Revolutionary Socialist Party - RSP	1	1534	I.N.D.I.A.
26	Shiromani Akali Dal - SAD	1	1584	OTHER
27	Sikkim Krantikari Morcha - SKM	1	1658	NDA
28	Samajwadi Party - SP	37	1680	I.N.D.I.A.
29	Telugu Desam - TDP	16	1745	NDA
30	Viduthalai Chiruthaigal Katchi - VCK	2	1847	I.N.D.I.A.
31	Yuva Jana Sramika Rythu Congress Party - YSR...	4	1888	OTHER
32	United People's Party, Liberal - UPPL	1	1998	OTHER
33	Apna Dal (Soneylal) - ADAL	1	2070	NDA
34	Rashtriya Loktantrik Party - RLTP	1	2484	I.N.D.I.A.
35	Zoram People's Movement - ZPM	1	2757	OTHER
36	Aazad Samaj Party (Kanshi Ram) - ASPKR	1	2989	OTHER
37	Lok Janshakti Party(Ram Vilas) - LJPRV	5	3165	NDA
38	Shiv Sena (Uddhav Balasaheb Thackrey) - SHS...	9	3369	I.N.D.I.A.
39	Voice of the People Party - VOTPP	1	3388	OTHER
40	Bharat Adivasi Party - BHRTADVSIP	1	3482	I.N.D.I.A.
41	Shiv Sena - SHS	7	3529	NDA
42	Nationalist Congress Party Sharadchandra Pa...	8	3620	I.N.D.I.A.

8. Which party alliance (NDA, I.N.D.I.A. or OTHER) won the most seats across all states?

```
SELECT
    p.party_alliance,
    COUNT(cr.Constituency_ID) AS Seats_Won
FROM constituencywise_results cr
JOIN partywise_results p ON cr.Party_ID = p.Party_ID
WHERE p.party_alliance IN ('NDA', 'I.N.D.I.A.', 'OTHER')
GROUP BY p.party_alliance
ORDER BY Seats_Won DESC;
```

Results Messages		
	party_alliance	Seats_Won
1	NDA	292
2	I.N.D.I.A.	234
3	OTHER	17

9. Winning candidates name, their party name, total votes, and the margin of victory for a specific state and constituency?

```
SELECT cr.Winning_Candidate, p.Party, p.party_alliance, cr.Total_Votes, cr.Margin, cr.Constituency_Name, s.State
FROM constituencywise_results cr
JOIN partywise_results p ON cr.Party_ID = p.Party_ID
JOIN statewide_results sr ON cr.Parliament_Constituency = sr.Parliament_Constituency
JOIN states s ON sr.State_ID = s.State_ID
WHERE s.State = 'Uttar Pradesh' AND cr.Constituency_Name = 'AMETHI';
```

Results		Messages					
	Winning_Candidate	Party	party_alliance	Total_Votes	Margin	Constituency_Name	State
1	KISHORI LAL	Indian National Congress - INC	I.N.D.I.A.	539228	167196	AMETHI	Uttar Pradesh

10. What is the distribution of EVM votes versus postal votes for candidates in a specific constituency?

```

SELECT
    cd.Candidate,
    cd.Party,
    cd.EVM_Votes,
    cd.Postal_Votes,
    cd.Total_Votes,
    cr.Constituency_Name
FROM constituencywise_details cd
JOIN constituencywise_results cr ON cd.Constituency_ID = cr.Constituency_ID
WHERE cr.Constituency_Name = 'MATHURA'
ORDER BY cd.Total_Votes DESC;

```

100 % No issues found

Results		Messages				
	Candidate	Party	EVM_Votes	Postal_Votes	Total_Votes	Constituency_Name
1	HEMAMALINI DHARMENDRA DEOL	Bharatiya Janata Party	507535	2529	510064	MATHURA
2	MUKESH DHANGAR	Indian National Congress	216043	614	216657	MATHURA
3	SURESH SINGH	Bahujan Samaj Party	188152	265	188417	MATHURA
4	BHANU PRATAP SINGH	Independent	15640	25	15665	MATHURA
5	NOTA	None of the Above	4527	36	4563	MATHURA
6	RAKESH KUMAR	Independent	4492	2	4494	MATHURA
7	JAGDISH PRASAD KAUSHIK ADVOCATE	Rashtriya Samta Vikas Party	3567	8	3575	MATHURA
8	RAVI VERMA	Independent	2724	0	2724	MATHURA
9	SHIKHA SHARMA	Independent	2246	5	2251	MATHURA
10	KAMAL KANT SHARMA	Independent	1491	12	1503	MATHURA
11	YOGESH KUMAR TALAN	Independent	1491	2	1493	MATHURA
12	MONI FALHARI BAPU	Independent	1322	4	1326	MATHURA
13	KSHETRA PAL SINGH	Independent	1206	5	1211	MATHURA
14	DR. RASHMI YADAV	Independent	1183	6	1189	MATHURA
15	PRAVESHANAND PURI	Independent	1081	4	1085	MATHURA
16	SURESH CHANDRA VAGHEL	Rashtriya Shoshit Samaj Party	822	7	829	MATHURA

11. Which parties won the most seats in s State, and how many seats did each party win?

```
SELECT
    p.Party,
    COUNT(cr.Constituency_ID) AS Seats_Won
FROM constituencywise_results cr
JOIN partywise_results p ON cr.Party_ID = p.Party_ID
JOIN statewise_results sr ON cr.Parliament_Constituency = sr.Parliament_Constituency
JOIN states s ON sr.State_ID = s.State_ID
WHERE s.state = 'Uttar Pradesh'
GROUP BY p.Party
ORDER BY Seats_Won DESC
```

100 % No issues found

Results Messages

	Party	Seats_Won
1	Samajwadi Party - SP	37
2	Bharatiya Janata Party - BJP	33
3	Indian National Congress - INC	6
4	Rashtriya Lok Dal - RLD	2
5	Aazad Samaj Party (Kanshi Ram) - ASPKR	1
6	Apna Dal (Soneylal) - ADAL	1

12. What are the total numbers of seats won by each party alliance (NDA, I.N.D.I.A. and OTHER) in each state for the India Elections 2024?

```
SELECT
    s.State AS State_Name,
    SUM(CASE WHEN p.party_alliance = 'NDA' THEN 1 ELSE 0 END) AS NDA_Seats_Won,
    SUM(CASE WHEN p.party_alliance = 'I.N.D.I.A.' THEN 1 ELSE 0 END) AS INDIA_Seats_Won,
    SUM(CASE WHEN p.party_alliance = 'OTHER' THEN 1 ELSE 0 END) AS OTHER_Seats_Won
FROM constituencywise_results cr
JOIN partywise_results p ON cr.Party_ID = p.Party_ID
JOIN statewise_results sr ON cr.Parliament_Constituency = sr.Parliament_Constituency
JOIN states s ON sr.State_ID = s.State_ID
WHERE p.party_alliance IN ('NDA', 'I.N.D.I.A.', 'OTHER')
GROUP BY s.State
ORDER BY s.State;
```

	State_Name	NDA_Seats_Won	INDIA_Seats_Won	OTHER_Seats_Won
1	Andaman & Nicobar Islands	1	0	0
2	Andhra Pradesh	21	0	4
3	Arunachal Pradesh	2	0	0
4	Assam	10	3	1
5	Bihar	30	9	1
6	Chandigarh	0	1	0
7	Chhattisgarh	10	1	0
8	Dadra & Nagar Haveli and Daman & Diu	1	0	1
9	Delhi	7	0	0
10	Goa	1	1	0
11	Gujarat	25	1	0
12	Haryana	5	5	0
13	Himachal Pradesh	4	0	0
14	Jammu and Kashmir	2	2	1
15	Jharkhand	9	5	0
16	Karnataka	19	9	0
17	Kerala	1	19	0
18	Ladakh	0	0	1
19	Lakshadweep	0	1	0
20	Madhya Pradesh	29	0	0
21	Maharashtra	17	30	1
22	Manipur	0	2	0
23	Meghalaya	0	1	1
24	Mizoram	0	0	1
25	Nagaland	0	1	0
26	Odisha	20	1	0
27	Puducherry	0	1	0
28	Punjab	0	10	3
29	Rajasthan	14	11	0
30	Sikkim	1	0	0
31	Tamil Nadu	0	39	0
32	Telangana	8	8	1
33	Tripura	2	0	0
34	Uttar Pradesh	36	43	1
35	Uttarakhand	5	0	0
36	West Bengal	12	30	0

13. Which candidate received the highest number of EVM votes of each constituency (Top 10)?

```

SELECT TOP 10
    cr.Constituency_Name,
    cd.Constituency_ID,
    cd.Candidate,
    cd.EVM_Votes
FROM constituencywise_details cd
JOIN constituencywise_results cr ON cd.Constituency_ID = cr.Constituency_ID
WHERE cd.EVM_Votes = (
    SELECT MAX(cd1.EVM_Votes)
    FROM constituencywise_details cd1
    WHERE cd1.Constituency_ID = cd.Constituency_ID)
ORDER BY cd.EVM_Votes DESC;

```

100 % ✓ No issues found

Results		Messages		
	Constituency_Name	Constituency_ID	Candidate	EVM_Votes
1	DHUBRI	S032	RAKIBUL HUSSAIN	1468549
2	INDORE	S1226	SHANKAR LALWANI	1223746
3	VIDISHA	S1218	SHIVRAJ SINGH CHOUHAN	1111556
4	BANGALORERURAL	S1023	DR C N MANJUNATH	1075553
5	RAIPUR	S268	BRIJMOHAN AGRAWAL	1047447
6	DIAMONDHARBOUR	S2521	ABHISHEK BANERJEE	1043493
7	NAVSARI	S0625	C R PATIL	1023366
8	GANDHINAGAR	S066	AMIT SHAH	999984
9	BANGALORENORTH	S1024	SHOBHA KARANDLAJE	982805
10	MALKAJGIRI	S297	EATALA RAJENDER	980712

14. Which candidate won and which candidate was the runner-up in each constituency of State for the 2024 elections?

```
WITH RankedCandidates AS (
SELECT
    cd.Constituency_ID,
    cd.Candidate,
    cd.Party,
    cd.EVM_Votes,
    cd.Postal_Votes,
    cd.EVM_Votes + cd.Postal_Votes AS Total_Votes,
    ROW_NUMBER() OVER (PARTITION BY cd.Constituency_ID ORDER BY cd.EVM_Votes + cd.Postal_Votes DESC) AS VoteRank
FROM constituencywise_details cd
JOIN constituencywise_results cr ON cd.Constituency_ID = cr.Constituency_ID
JOIN statewide_results sr ON cr.Parliament_Constituency = sr.Parliament_Constituency
JOIN states s ON sr.State_ID = s.State_ID
WHERE s.State = 'Maharashtra')
SELECT cr.Constituency_Name,
MAX(CASE WHEN rc.VoteRank = 1 THEN rc.Candidate END) AS Winning_Candidate,
MAX(CASE WHEN rc.VoteRank = 2 THEN rc.Candidate END) AS Runnerup_Candidate
FROM RankedCandidates rc
JOIN constituencywise_results cr ON rc.Constituency_ID = cr.Constituency_ID
GROUP BY cr.Constituency_Name
ORDER BY cr.Constituency_Name;
```

100 % ✓ No issues found

Results Messages

	Constituency_Name	Winning_Candidate	Runnerup_Candidate
1	AHMEDNAGAR	NILESH DNYANDEV LANKE	DR. SUJAY RADHAKRISHNA VIKHEPA
2	AKOLA	ANUP SANJAY DHOTRE	ABHAY KASHINATH PATIL
3	AMRAVATI	BALWANT BASWANT WANKHA...	NAVNEET RAVI RANA
4	AURANGABAD	BHUMARE SANDIPANRAO ASA...	IMTIAZ JALEEL SYED
5	BARAMATI	SUPRIYA SULE	SUNETRA AJITDADA PAWAR
6	BEED	BAJRANG MANOHAR SONWANE	PANKAJA GOPINATHRAO MUNDE
7	BHANDARAGONDI...	DR. PRASHANT YADAORAO PA...	SUNIL BABURAO MENDHE
8	BHIWANDI	BALYA MAMA - SURESH GOPIN...	KAPIL MORESHWAR PATIL
9	BULDHANA	JADHAV PRATAPRAO GANPAT...	NARENDRA DAGDU KHEDEKAR
10	CHANDRAPUR	DHANORKAR PRATIBHA SURE...	MUNGANTIWAR SUDHIR SACCCHIDANA
11	DHULE	BACHHAV SHOBHA DINESH	BHAMRE SUBHASH RAMRAO
12	DINDORI	BHASKAR MURLIDHAR BHAGARE	DR. BHARATI PRAVIN PAWAR
13	GADCHIROLI-CHI...	DR. KIRSAN NAMDEO	ASHOK MAHADEORAO NETE
14	HATKANANGALE	DHAIRYASHEEL SAMBHAJIRAO...	SATYAJEET BABASAHEB PATIL (AABA
15	HINGOLI	AASHTIKAR PATIL NAGESH BA...	BABURAO KADAM KOHALIKAR
16	JALGAON	SMITA UDAY WAGH	KARAN BALASAHEB PATIL - PAWAR
17	JALNA	KALYAN VAJJINATHRAO KALE	DANVE RAOSAHEB DADARAO
18	KALYAN	DR. SHRIKANT EKNATH SHINDE	VAISHALI DAREKAR - RANE
19	KOLHAPUR	CHHATRAPATI SHAHU SHAHAJI	SANJAY SADASHIVRAO MANDLIK
20	LATUR	DR. KALGE SHIVAJI BANDAPPA	SUDHAKAR TUKARAM SHRANGARE
21	MADHA	MOHITE-PATIL DHAIRYASHEEL ...	RANJEETSINGH HINDURAO NAIK NIME
22	MAVAL	SHRIRANG APPA CHANDU BAR...	SANJOG BHIKU WAGHERE PATIL
23	MUMBAINORTH	PIYUSH GOYAL	BHUSHAN PATIL
24	MUMBAINORTHCE...	GAIKWAD VARSHA EKNATH	ADV UJWAL NIKAM

25	MUMBAINORTHEA...	SANJAY DINA PATIL	MIHIR CHANDRAKANT KOTECHEA
26	MUMBAINORTHW...	RAVINDRA DATTARAM WAIKAR	AMOL GAJANAN KIRTIKAR
27	MUMBAISOUTH	ARVIND GANPAT SAWANT	YAMINI YASHWANT JADHAV
28	MUMBAISOUTHCE...	ANIL YESHWANT DESAI	RAHUL RAMESH SHEWALE
29	NAGPUR	NITIN JAIRAM GADKARI	VIKAS THAKRE
30	NANDED	CHAVAN VASANTRAO BALWAN...	CHIKHALIKAR PRATAPRAO GOVINDRAO
31	NANDURBAR	ADV GOWAAL KAGADA PADAVI	DR. HEENA VIJAYKUMAR GAVIT
32	NASHIK	RAJABHAU (PARAG) PRAKASH ...	GODSE HEMANT TUKARAM
33	OSMANABAD	OMPRAKASH BHUPALSINH ALI...	ARCHANA RANA JAGJITSINH PATIL
34	PALGHAR	DR. HEMANT VISHNU SAVARA	BHARTI BHARAT KAMDIA
35	PARBHANI	JADHAV SANJAY (BANDU) HA...	JANKAR MAHADEV JAGANNATH
36	PUNE	MURLIDHAR MOHOL	DHANGEKAR RAVINDRA HEMRAJ
37	RAIGAD	TATKARE SUNIL DATTATREY	ANANT GEETE
38	RAMTEK	Shyamkumar (Babelu) Daulat Ba...	RAJU DEONATH PARVE
39	RATNAGIRI-SINDH...	NARAYAN TATU RANE	VINAYAK BHABURAO RAUT
40	RAVER	KHADSE RAKSHA NIKHIL	SHRIRAM DAYARAM PATIL
41	SANGLI	VISHAL (DADA) PRAKASHBAPU...	SANJAY (KAKA) PATIL
42	SATARA	SHRIMANT CHH UDAYANRAJE ...	SHASHIKANT JAYVANTRAO SHINDE
43	SHIRDI	BHAUSAHEB RAJARAM WAKCH...	LOKHANDE SADASHIV KISAN
44	SHIRUR	DR. AMOL RAMSING KOLHE	ADHALRAO SHIVAJI DATTATREY
45	SOLAPUR	PRANITI SUSHILKUMAR SHINDE	RAM VITTHAL SATPUTE
46	THANE	NARESH GANPAT MHASKE	RAJAN BABURAO VICHARE
47	WARDHA	AMAR SHARADRAO KALE	RAMDAS CHANDRABHAN TADAS
48	YAVATMAL-WASHIM	SANJAY UTTAMRAO DESHMUKH	RAJSHRITAI HEMANT PATIL (MAHALL)

Query Explanation:

1. Common Table Expression (CTE) RankedCandidates

- **WITH RankedCandidates AS:** This section creates a Common Table Expression (CTE) named RankedCandidates, which is used to rank candidates based on their total votes within each constituency.
- **SELECT ... FROM constituencywise_details cd:** This part of the CTE retrieves data from the constituencywise_details table, including:
 - cd.Constituency_ID: The ID of the constituency.
 - cd.Candidate: The name of the candidate.
 - cd.Party: The party of the candidate.
 - cd.EVM_Votes: The number of EVM votes received by the candidate.
 - cd.Postal_Votes: The number of postal votes received by the candidate.
 - cd.EVM_Votes + cd.Postal_Votes AS Total_Votes: The total votes received by the candidate (sum of EVM and postal votes).
- **ROW_NUMBER() OVER (PARTITION BY cd.Constituency_ID ORDER BY cd.EVM_Votes + cd.Postal_Votes DESC) AS VoteRank:**
 - ROW_NUMBER() is a window function that assigns a unique rank to each candidate within their respective constituency.
 - PARTITION BY cd.Constituency_ID ensures that the ranking is done separately for each constituency.
 - ORDER BY cd.EVM_Votes + cd.Postal_Votes DESC orders candidates by their total votes in descending order, so the candidate with the most votes gets rank 1.
- **JOIN ... ON ...:** Joins are performed to combine information from related tables:
 - constituencywise_results cr ON cd.Constituency_ID = cr.Constituency_ID

- statewide_results sr ON cr.Parliament_Constituency = sr.Parliament_Constituency
- states s ON sr.State_ID = s.State_ID
- **WHERE s.State = 'Maharashtra':** Filters the results to include only data related to the state of Maharashtra.

2. Main Query

- **SELECT ... FROM RankedCandidates rc:** The main query retrieves data from the RankedCandidates CTE to find the winning and runner-up candidates for each constituency.
- **MAX(CASE WHEN rc.VoteRank = 1 THEN rc.Candidate END) AS Winning_Candidate:**
 - CASE WHEN rc.VoteRank = 1 THEN rc.Candidate END selects the candidate with the highest vote rank (rank 1) for each constituency.
 - MAX(...) ensures that only one candidate is selected for each constituency.
- **MAX(CASE WHEN rc.VoteRank = 2 THEN rc.Candidate END) AS Runnerup_Candidate:**
 - CASE WHEN rc.VoteRank = 2 THEN rc.Candidate END selects the candidate with the second highest vote rank (rank 2) for each constituency.
 - MAX(...) ensures that only one candidate is selected for each constituency.
- **JOIN ... ON ...:** Joins the RankedCandidates CTE with the constituencywise_results table to get the constituency names.
- **GROUP BY cr.Constituency_Name:** Groups the results by constituency name to ensure that the winning and runner-up candidates are determined for each constituency.
- **ORDER BY cr.Constituency_Name:** Orders the results alphabetically by constituency name for easier readability.

15. For the state of Maharashtra, what are the total number of seats, total number of candidates, total number of parties, total votes (including EVM and postal), and the breakdown of EVM and postal votes?

```
SELECT
COUNT(DISTINCT cr.Constituency_ID) AS Total_Seats,
COUNT(DISTINCT cd.Candidate) AS Total_Candidates,
COUNT(DISTINCT p.Party) AS Total_Parties,
SUM(cd.EVM_Votes + cd.Postal_Votes) AS Total_Votes,
SUM(cd.EVM_Votes) AS Total_EVM_Votes,
SUM(cd.Postal_Votes) AS Total_Postal_Votes
FROM constituencywise_results cr
JOIN constituencywise_details cd ON cr.Constituency_ID = cd.Constituency_ID
JOIN statewide_results sr ON cr.Parliament_Constituency = sr.Parliament_Constituency
JOIN states s ON sr.State_ID = s.State_ID
JOIN partywise_results p ON cr.Party_ID = p.Party_ID
WHERE s.State = 'Maharashtra'
```

100 % ✓ No issues found

Results Messages

	Total_Seats	Total_Candidates	Total_Parties	Total_Votes	Total_EVM_Votes	Total_Postal_Votes
1	48	1114	7	57179133	56969710	209423